

Zuobin Xiong

Assistant Professor, Computer Science

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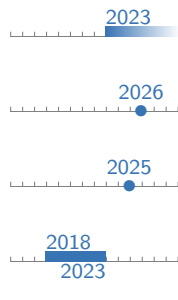
in [Zuobin-Xiong](#)

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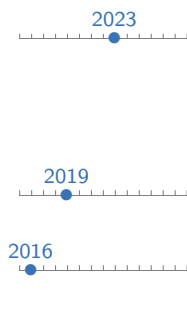
May the Force be with you.

STAR WARS 1977

Experience

- 
- Tenure-track Assistant Professor**, *Department of Computer Science*.
University of Nevada, Las Vegas, Las Vegas, NV
 - Visiting Scholar**, *Thomas Lord Department of Computer Science*.
2026.06-2026.08, University of Southern California, Los Angeles, CA
 - Visiting Scholar**, *Thomas Lord Department of Computer Science*.
2025.07-2025.08, University of Southern California, Los Angeles, CA
 - Graduate Research/Teaching Assistant**, *Department of Computer Science*.
Georgia State University, Atlanta, GA

Education

- 
- Ph.D.**, *in Computer Science*.
Georgia State University, Atlanta, USA
 - Advisors: Dr. Wei Li & Dr. Zhipeng Cai
 - Dissertation: Towards Privacy Preservation of Federated Learning in Artificial Intelligence of Things
 - M.E.**, *in Computer Science & Technology*.
Harbin Engineering University, Harbin, China
 - B.S.**, *in Mathematics & Applied Mathematics*.
Northeast Forestry University, Harbin, China

Research Area

- AI/ML
 - Distributed Machine Learning
 - Generative Models
 - Machine Unlearning
 - AI4Science
- Data Privacy
 - Differential Privacy
 - Private Data Publishing
- Data Mining
 - Trajectory Data Mining
 - Graph Data Mining/Learning
- the Internet of Things
 - Edge Computing
 - Artificial Intelligence of Things (AIoT)
 - Social Internet of Things (SIoT)

Honors & Awards

2024	UNLV University Faculty Travel Grant , <i>Las Vegas, NV</i> . November. Awarded by University Faculty Travel Committee (UFTC)
2024	NSF SaTC Aspiring PI Workshop Travel Grant , <i>Chicago, Illinois</i> . April. Awarded by NSF 2024 SaTC Aspiring PI Workshop
2023	Outstanding Graduate Student Award , <i>Atlanta, Georgia</i> . April. Awarded by the College of Arts & Sciences, Georgia State University
2023	AAAI Student Travel Scholarship , <i>D.C., USA</i> . February. Awarded by AAAI -23 and the Association for the Advancement of Artificial Intelligence
2022	Best Paper Award , <i>Espoo, Finland</i> . August. Awarded by the 8th IEEE International Conference on Smart Data (SmartData 2022)
2021	Outstanding Graduate Research Award , <i>Atlanta, Georgia</i> . April. Awarded by the Department of Computer Science, Georgia State University

Fundings

2026 2029	NSF REU , <i>Hands-On Research in Federated Learning Security through Red Team vs. Blue Team Exercises</i> . Co-PI, \$460,922
2025 2027	NSF HSI , <i>A Guided Pathway to Enhancing HSI Student Experience and Success in Generative AI with the Planting of Education Oriented GPU Cluster</i> . PI, \$199,413
2025 2025	NSF RII Seed Grant , <i>Harnessing Data Revolution for Fire Science (HDRFS) Data Analytics Mentor</i> . PI, \$4,750
2025 2027	NSF Research Fellows , <i>An Explainable AI-Supported Performance Monitoring System in Distributed Sustainable Energy Networks</i> . Single PI, \$286,889
2024 2026	UNLV Faculty Opportunity Award , <i>Are You Ready for College? An Explainable AI-Supported Efficient Solution for College Students Mental Health Condition Detection and Beyond</i> . PI, \$35,000
2024 2026	South Korea IITP , <i>Intelligent IoT Security: Next-Generation Cyber Defense Mechanisms and Vulnerability Exploration Using Language Models</i> . Co-PI, \$357,739
2023 2025	NSF RII Track-2 FEC Seed Grant , <i>AI SUSTEIN Grant Program</i> . PI, \$30,000

Journal Papers

- [1] Z. Zhang, Y. Wang, H. Teng, W. Hou, Z. Xiong, and Z. Cai, "Co-UPP: A co-optimized UAV path planning framework for balancing truth discovery and environmental sensing in mobile crowdSensing", *IEEE Transactions on Vehicular Technology*, 2026.
- [2] Y. Nam, D. Lim, S. Im, Z. Xiong, and J. Park, "FAIRITHM: Unlearning bias and rewriting for fairness", *International Journal of Human-Computer Interaction*, 2026.
- [3] D. Davidson, M. Jackson, R. Hussain, Z. Xiong, and J. Son, "Machine learning over encrypted data: A survey of fully homomorphic encryption for AI systems", *Artificial Intelligence Review*, 2026.

- [4] H. Jeong*, Z. Xiong, H. Kang, K. Kang, and J. Son, "Fine-grained fashion classification with limited labeled data via generative augmentation", *Intelligent and Converged Networks*, vol. 6, no. 4, pp. 311–325, 2025.
- [5] Z. Xiong, W. Li, Y. Li, and Z. Cai, "Distributed generative model: A data synthesizing framework for multi-source heterogeneous data", *IEEE Transactions on Artificial Intelligence*, pp. 1–1, 2025. DOI: 10.1109/TAI.2025.3575548.
- [6] A. Huang*, Z. Cai, and Z. Xiong, "A survey of machine unlearning in generative ai models: Methods, applications, security, and challenges", *IEEE Internet of Things Journal*, pp. 1–1, 2025. DOI: 10.1109/JIOT.2025.3570989.
- [7] J. Choi, Z. Xiong, and K. Kang, "Long short-term memory-based computerized numerical control machining center failure prediction model", *Mathematics*, vol. 13, no. 7, p. 1093, 2025.
- [8] L. Zhang, S. Park, Z. Xiong, J. Son, and Y. Lee, "Understanding illicit promotional contents on short video platforms", *Tsinghua Science and Technology*, 2024.
- [9] G. Lu, Z. Xiong, R. Li, N. Mohammad, Y. Li, and W. Li, "Defeat: A decentralized federated learning against gradient attacks", *High-Confidence Computing*, vol. 3, no. 3, p. 100 128, 2023.
- [10] L. Jiang, Y. Yan, Z. Tian, Z. Xiong, and Q. Han, "Personalized sampling graph collection with local differential privacy for link prediction", *World wide web*, vol. 26, no. 5, pp. 2669–2689, 2023.
- [11] Z. Xiong, "Towards privacy preservation of federated learning in artificial intelligence of things", 2023.
- [12] Z. Xiong, Z. Cai, C. Hu, D. Takabi, and W. Li, "Towards neural network-based communication system: Attack and defense", *IEEE Transactions on Dependable and Secure Computing*, vol. 20, no. 4, pp. 3238–3250, 2022.
- [13] J. Wang, Z. Xiong, Q. Han, X. Han, and D. Yang, "Top-k socially constrained spatial keyword search in large siot networks", *IEEE Internet of Things Journal*, vol. 9, no. 12, pp. 9280–9289, 2021.
- [14] G. Li, G. Yin, Z. Xiong, and F. Chen, "Cgpp-poi: A recommendation model based on privacy protection", *Wireless Communications and Mobile Computing*, vol. 2021, no. 1, p. 4873574, 2021.
- [15] Z. Xiong, Z. Cai, D. Takabi, and W. Li, "Privacy threat and defense for federated learning with non-iid data in aiot", *IEEE Transactions on Industrial Informatics*, vol. 18, no. 2, pp. 1310–1321, 2021.
- [16] Z. Cai, Z. Xiong, H. Xu, P. Wang, W. Li, and Y. Pan, "Generative adversarial networks: A survey toward private and secure applications", *ACM Computing Surveys (CSUR)*, vol. 54, no. 6, pp. 1–38, 2021.
- [17] X. Tao, Y. Peng, F. Zhao, *et al.*, "Gated recurrent unit-based parallel network traffic anomaly detection using subagging ensembles", *Ad Hoc Networks*, vol. 116, p. 102465, 2021.
- [18] Z. Xiong, H. Xu, W. Li, and Z. Cai, "Multi-source adversarial sample attack on autonomous vehicles", *IEEE Transactions on Vehicular Technology*, vol. 70, no. 3, pp. 2822–2835, 2021.
- [19] Z. Xiong, Z. Cai, Q. Han, A. Alrawais, and W. Li, "Adgan: Protect your location privacy in camera data of auto-driving vehicles", *IEEE Transactions on Industrial Informatics*, vol. 17, no. 9, pp. 6200–6210, 2020.
- [20] Q. Han, Z. Xiong, and K. Zhang, "Research on trajectory data releasing method via differential privacy based on spatial partition", *Security and Communication Networks*, vol. 2018, no. 1, p. 4248092, 2018.

Conference Papers

- [21] C. Dahal, A. Balasubramaniam, and Z. Xiong, "GONE: Structural knowledge unlearning via neighborhood-expanded distribution shaping", in *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Budapest, Hungary: Association for Computational Linguistics, 2026.
- [22] F. Jiang, Z. Xiong, A. Huang, Z. Cai, and Y. Li, "Membership is ownership: A robust ownership verification framework for diffusion models", in *2026 IEEE International Conference on Data Mining (ICDM)*, IEEE, 2026.
- [23] D. Davidson, M. Kang, Y. Lee, Z. Xiong, and J. Son, "Gpu accelerated fully homomorphic encryption framework for end-to-end logistic regression", in *International Conference on Security and Privacy in Communication Systems*, Springer, 2026, pp. 449–467.
- [24] A. Huang, J. Son, and Z. Xiong, "Watch your step: Information injection in diffusion models via shadow timestep embedding", in *International Conference on Machine Learning (ICML)*, PMLR, 2026, pp. 1–8.
- [25] Z. Xiong, D. Mukherjee, H. Cho, and W. Li, "Rethinking the transferable adversarial attacks and robust defense in federated learning", in *2026 International Conference on Computer Communications and Networks (ICCCN)*, IEEE, 2026, pp. 1–8.
- [26] S. Nakarmi*, J. Son, Y. Zhao, and Z. Xiong, "Fed-listing: Federated label distribution inference in graph neural networks", in *2026 International Conference on Computer Communications and Networks (ICCCN)*, IEEE, 2026, pp. 1–8.
- [27] G. C. Amaizu, A. M. V. V. Sai, Z. Xiong, and Y. Li, "Trusted medical ai: Blockchain-backed device authentication with digital twin-enhanced xai for lung cancer detection", in *2025 IEEE International Performance, Computing, and Communications Conference (IPCCC)*, IEEE, 2025, pp. 1–8.
- [28] A. Huang*, Z. Xiong, M. Ye, and J. Son, "It is hard to unlearn dogged backdoor samples in diffusion models", in *NeurIPS 2025 Workshop: Reliable ML from Unreliable Data*.
- [29] C. Dahal* and Z. Xiong, "How well do llms unlearn facts?-a knowledge graph perspective", in *Women in Machine Learning Workshop@ NeurIPS 2025*.
- [30] G. C. Amaizu, A. M. Sai, Y. Li, and Z. Xiong, "Trusted medical ai: Blockchain-backed device authentication with digital twin-enhanced xai for lung cancer detection", in *Proceedings of the IEEE International Performance Computing and Communications Conference (IPCCC)*, November 21–23, 2025, Austin, TX, USA, Nov. 2025.
- [31] J. Ahn*, Z. Xiong, H. Cho, K. Kang, and J. Son, "Efficient phishing website detection via html tag sequence analysis using encoder models", in *International Conference on Computer Communications and Networks*, IEEE, 2025.
- [32] A. Huang*, J. Son, and Z. Xiong, "Ddsnet: A lightweight dense depthwise separable network for tumor classification", in *ACM/SIGAPP Symposium on Applied Computing*, ACM, 2025.
- [33] L. Wang*, Z. Xiong, G. Luo, W. Li, and A. Chen, "Fcfl: A fairness compensation-based federated learning scheme with accumulated queues", in *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, Springer, 2024, pp. 386–402.
- [34] Z. Xiong, W. Li, and Z. Cai, "Appro-fun: Approximate machine unlearning in federated setting", in *2024 33rd International Conference on Computer Communications and Networks (ICCCN)*, IEEE, 2024, pp. 1–9.
- [35] Z. Xiong, W. Li, Y. Li, and Z. Cai, "Exact-fun: An exact and efficient federated unlearning approach", in *2023 IEEE International Conference on Data Mining (ICDM)*, IEEE, 2023, pp. 1439–1444.
- [36] H. Xu, Z. Cai, Z. Xiong, and W. Li, "Backdoor attack on 3d grey image segmentation", in *2023 IEEE International Conference on Data Mining (ICDM)*, IEEE, 2023, pp. 708–717.
- [37] L. Li, W. Liu, Z. Xiong, and Y. Wang, "Sequence-based modeling for temporal knowledge graph link prediction", in *International Conference on Artificial Neural Networks*, Springer, 2023, pp. 550–562.

- [38] Z. Xiong, W. Li, and Z. Cai, "Federated generative model on multi-source heterogeneous data in iot", in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, 2023, pp. 10 537–10 545.
- [39] G. Lu, Z. Xiong, R. Li, and W. Li, "Decentralized federated learning: A defense against gradient inversion attack", in *International Wireless Internet Conference*, Springer, 2022, pp. 44–56.
- [40] G. Lu, Z. Xiong, J. Meng, and W. Li, "Pairwise gaussian graph convolutional networks: Defense against graph adversarial attack", in *GLOBECOM 2022-2022 IEEE Global Communications Conference*, IEEE, 2022, pp. 4371–4376.
- [41] B. Xie, H. Xu, Z. Xiong, Y. Li, and Z. Cai, "A self-supervised purification mechanism for adversarial samples", in *2022 IEEE International Conferences on Internet of Things (iThings) and IEEE Green Computing & Communications (GreenCom) and IEEE Cyber, Physical & Social Computing (CPSCom) and IEEE Smart Data (SmartData) and IEEE Congress on Cybermatics (Cybermatics)*, IEEE, 2022, pp. 501–509.
- [42] L. Li, S. Hu, J. Chen, Y. Wang, and Z. Xiong, "Exp-softlexicon lattice model integrating radical-level features for chinese ner.", in *The 34th International Conference on Software Engineering & Knowledge Engineering*, 20122, pp. 329–334.
- [43] Z. Xiong, W. Li, Q. Han, and Z. Cai, "Privacy-preserving auto-driving: A gan-based approach to protect vehicular camera data", in *2019 IEEE International Conference on Data Mining (ICDM)*, IEEE, 2019, pp. 668–677.

--- Presentations

Conference Presentations

- 
Appro-fun: Approximate Machine Unlearning in Federated Setting, *Big Island, Hawaii*.
 July. International Conference on Computer Communications and Networks
- 
Exact-Fun: An Exact and Efficient Federated Unlearning Approach, *Beijing, China*.
 December. IEEE International Conference on Data Mining
- 
Privacy-Preserving Auto-Driving: a GAN-based Approach to Protect Vehicular Camera Data, *Beijing, China*.
 June. IEEE International Conference on Data Mining

Invited Talks

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Towards Privacy Preservation of Federated Learning in Artificial Intelligence of Things, *Online*.
 September. Department of Electrical and Computer Engineering, Virginia Commonwealth University
- 
Privacy Threats and Defense in Federated Learning, *Online*.
 August. University of Electronic Science and Technology of China

--- Teaching

Teaching at UNLV

- 
Fall: CS422/622 Intro to Machine Learning, *Enrollment 24*, SEI, TBD.
 Instructor. Under/Graduate Level
- 
Spring: CS302 Data Structures, *Enrollment 22*, SEI, TBD.
 Instructor. Undergraduate Level
- 
Fall: CS789 Advanced Topics in Computer Science, *Enrollment 10*, SEI, 4.58/5.0.
 Instructor. Graduate Level



Spring: CS302 Data Structures, *Enrollment 40*, SEI, 4.06/5.0.

Instructor. Undergraduate Level



Fall: CS789 Advanced Topics in Computer Science, *Enrollment 10*, SEI, 4.72/5.0.

Instructor. Graduate Level

Teaching at GSU



Spring: CSC4222/6222 Intro to Cybersecurity, *Enrollment 55*, SEI, 4.5/5.0.

Instructor. Under/Graduate Level



Fall: CSC4520/6520 Design and Analysis of Algorithms, *Enrollment 57*, SEI, 4.7/5.0.

Instructor. Under/Graduate Level

Student Mentoring

Current Ph.D. Students



Suprim Nakarmi, *Ph.D. Student*, Advising.

University of Nevada, Las Vegas



Chahana Dahal, *Ph.D. Student*, Advising.

University of Nevada, Las Vegas



An Huang, *Ph.D. Student*, Advising.

University of Nevada, Las Vegas



Hongbi Jeong, *Ph.D. Student*, Co-advising with Dr. Junggab Son.

University of Nevada, Las Vegas

Previous Students



Eunyoung Jang, *M.S.*, Computer Science.

University of Nevada, Las Vegas



Syed Shariq Ahmed, *M.S.*, Computer Science.

University of Nevada, Las Vegas

Dissertation Committee



Chol Park, *Ph.D.*, Computer Science.

University of Nevada, Las Vegas



Austin Janushan, *M.S.*, Computer Science.

University of Nevada, Las Vegas

Professional Services

Editorship

Associate Editor SN Computer Science, 2026 - Present

Associate Editor IEEE Transactions on Neural Networks and Learning Systems (TNNLS), 2026 - Present

Associate Editor High-Confidence Computing, 2025 - Present

Associate Editor IEEE Networking Letters, 2024 - Present

Guest Editor MDPI Electronics

PC Member

18th International Conference on Wireless Artificial Intelligent Computing Systems and Applications (WASA)

32nd International Conference on Artificial Neural Networks (ICANN)

1st International Conference on Meta Computing (ICMC)

Reviewer

Conference IEEE Global Communications Conference (GLOBECOM)

IEEE Cyber Science and Technology Congress

ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)

Journal ACM Transactions on Sensor Networks (TOSN)

Discrete Mathematics, Algorithms and Applications (DMAA)

Elsevier Neurocomputing

Elsevier Computer & Security

Elsevier Computer Communications

Elsevier High-Confidence Computing

IEEE Transactions on Industrial Informatics (TII)

IEEE Transactions on Vehicular Technology (TVT)

IEEE Internet of Things Journal (IoT-J)

IEEE Transactions on Wireless Communications (TWC)

IEEE Transactions on Knowledge and Data Engineering (TKDE)

IEEE Transactions on Network Science and Engineering (TNSE)

IEEE Transactions on Computational Social Systems (TCSS)

IEEE Networking Letters

Security and Communication Networks

Springer Scientific Reports - Nature

Springer Artificial Intelligence Review

Springer Journal of Big Data

Academic Membership

Member IEEE

AAAI

Languages

Chinese Native

English Proficient

Cantonese Fluent

Mother Tongue

Presentations and Lectures given in English

Occasional practice with TVB